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BUILT FOR SPRINTS, NEEDED FOR MARATHONS: MAPPING THE GAP BETWEEN SCIENCE, USERS, AND IMPLEMENTATION IN DIGITAL HABIT FORMATION APPS

Completed Research Paper

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Abstract

Despite the proliferation of habit formation apps, it remains unclear which design principles are perceived as most important by stakeholders. This study investigates the perceived importance of 18 design principles for digital habit formation and their implementation in commercial apps. We collected data from two stakeholder groups (22 habit-formation researchers and 108 app users) and combined the data with related research that analyzed feature implementation across 57 commercial habit apps. Our findings reveal strong consensus on pragmatic design, while theoretically critical mechanisms showed significant divergence: scientists valued Contextual Cues ($p = 0.016$), Social features ($p = 0.019$), and Fading Reminders ($p = 0.048$) more highly than users did. Key automaticity-related features were nearly absent in commercial apps, suggesting that while these apps support initial action, they fail to foster lasting habits. These findings underscore the need for evidence-based design that facilitates a transition from external regulation to automatic execution of behaviors.

Keywords: Habit Formation, Behavior Change Intervention, Design Principle, Disease Prevention

1 Introduction

Non-communicable diseases (NCDs) such as cardiovascular conditions, diabetes, and mental health disorders are the leading causes of illness and death worldwide, accounting for more than 70% of global deaths (WHO, 2013, 2022). Many of these diseases are linked to modifiable lifestyle behaviors, including physical activity, sleep quality, and stress management. Although improving these behaviors is central to prevention and treatment, individuals struggle to translate their intentions into behavior, a phenomenon referred to as the intention-behavior gap (Faries, 2016; Orbell & Sheeran, 1998). Habits help bridge this gap by transforming behaviors into automatic responses (Lally et al., 2010; Stark et al., 2023; Verplanken, 2018). Habits form when a behavior is repeated in the same context until it becomes an automatic response to a cue (Lally et al., 2010; Verplanken, 2018). For instance, someone who takes a walk every morning after coffee may eventually do so without thinking about it. The sight of the coffee cup becomes the cue to walk. Once formed, habits enable us to act with minimal conscious effort, freeing mental resources and promoting consistency (Bargh, 1994; Gardner et al., 2012).

In recent years, apps have become popular tools for building and maintaining habits. Thousands of habit apps are available via the Apple App Store and Google Play Store, promising to help users establish routines for physical activity, sleep hygiene, or productivity (Reinsch, Kählig, et al., 2025). These apps offer engagement-focused features, such as reminders, progress tracking, and gamification, to repeat the behavior. Yet, habit app features are often derived from product design conventions rather than behavioral science principles (Stawarz et al., 2015; Zhu et al., 2024), without a clear focus on forming and strengthening habits and, hence, achieving automaticity (Stark et al., 2023). This creates a tension

in the design of habit-formation apps. While product logic often prioritizes continued app engagement, habit formation theory emphasizes the gradual transition from externally supported action to context-triggered automatic behavior. Particularly in the context of NCDs, meaningful health outcomes depend on sustained lifestyle change rather than short-term engagement alone (Stark et al., 2023).

Early research in human–computer interaction and behavioral design provided initial insights into how apps can support habit formation. Stawarz et al. (2014) found that most commercial apps relied on time-based notifications, neglecting the routines and contextual cues essential for developing habits. In a subsequent study, Stawarz et al. (2015) demonstrated that reminders, although effective for prompting behavioral repetition, paradoxically impede the development of automaticity by maintaining external dependency rather than fostering context-driven habit formation. In contrast, event-based cues linked to daily routines proved more effective for lasting habit formation. Yet, most apps continue to prioritize self-tracking and motivation features over contextual embedding. A recent approach to supporting theory-informed app development was introduced by Stark et al. (2023), who identified thirteen design principles (DPs) for digital habit formation through a design-oriented synthesis of psychological theory and a content analysis of 57 commercially available habit apps. In doing so, they integrated behavioral and technological mechanisms spanning all stages of habit formation, from intention-formulation to automaticity, into a structured design space for digital habit support. While the DPs to support habit formation mark a conceptual advance, a major challenge remains. The relative importance of the individual DPs has not yet been empirically assessed. As a result, developers and researchers still lack evidence on which principles are most crucial for forming habits.

This paper builds on Stark et al. (2023) by aiming to classify and prioritize DPs for digital habit formation. By identifying which DPs are considered necessary by different stakeholder groups, this study captures two analytically distinct yet complementary perspectives on the design of digital habit formation. Scientists contribute theory-informed judgments grounded in behavioral and habit-formation research. End users, in contrast, contribute experience-informed judgments shaped by everyday app use. Comparing these perspectives allows us to assess whether design principles that are theoretically central to sustained habit formation are also perceived as relevant by those expected to engage with such systems in practice. Without clarity on which DP genuinely drives habit formation, scientists risk refining an overly broad or even conflicting set of recommendations. At the same time, practitioners continue to design apps that focus on engagement rather than lasting behavior change. Clarifying these priorities helps bridge the gap between theoretical frameworks and practical design. Accordingly, this study addresses the following research questions (RQ), aimed at examining how theory-informed design principles are perceived and translated into practice across two selected stakeholder perspectives:

RQ1: What is the perceived importance of digital habit formation design principles among scientific experts and end-users?

Understanding stakeholder priorities alone, however, is insufficient without examining their translation into practice, which is why we further integrate the results of RQ1 with data available in related research about the integration of DPs into commercially available habit apps as investigated in (Stark et al., 2023), and hence answer RQ2:

RQ2: To what extent do commercial habit formation applications implement design principles aligned with stakeholder-perceived importance?

To address the first RQ, we conducted a cross-sectional survey among two key stakeholder groups: (1) individuals engaged in habit formation, hereafter referred to as habit app users, and (2) scientists engaged in habit formation and digital health. By comparing their perspectives, we identify areas of convergence and divergence, revealing which design elements are valued consistently and which remain contested. To answer RQ2, we integrated the developer perspective by comparing our findings with the implementation data reported by Stark et al. (2023), who analyzed 57 habit apps, to determine the extent to which DPs are reflected in real-world applications. This triangulated approach enables us to contrast perceived importance (as reported by users and scientists) with implementation (by developers), thereby highlighting design gaps and priority areas for evidence-based development of habit apps. Theoretically, this study provides a foundation for empirically examining and refining individual DPs, thereby

contributing to a more evidence-based understanding of digital habit formation. For practice, it offers guidance for app developers to focus on impactful features that facilitate sustainable and automatic behavior change.

2 Theoretical Background

Habits are automatic behavioral responses that develop through repeated actions in stable contexts, where contextual cues (e.g., time, place, or preceding activities) consistently trigger the behavior (Lally et al., 2010; Verplanken, 2018). Examples for cues include brushing one's teeth every evening before bed (time) or stretching after finishing work (preceding action). Habits are crucial in enabling individuals to act efficiently and consistently in daily life. Once established, they operate with minimal conscious control, allowing people to perform routine actions without deliberate thought or decision-making (Wood & Neal, 2007). This cognitive automation frees up mental resources for demanding tasks and reduces the effort required for self-regulation (Verplanken, 2018). Thus, habits support behavioral stability, making it more likely that goal-directed actions, such as exercising or taking medication, are carried out regularly even in the absence of strong motivation (Lally et al., 2010). Habit formation is conceptualized into four stages (Lally et al., 2010; Lally & Gardner, 2013):

In the first stage, **Decision to Act**, individuals form an intention to change or establish a habit. In this stage, external support can help define an implementation intention (e.g., "If I finish dinner, then I will go for a short walk"). The second stage, **Translation of Intention into Action**, converts the implementation intention into action. Digital interventions can assist through reminders and prompts that make execution easier and reduce the intention-behavior gap. In the third stage, **Repetition and Strengthening**, the behavior is repeatedly performed in response to contextual cues. Consistency strengthens the cue-behavior link, and support mechanisms, such as feedback and reminders that help remember the implementation intention (e.g., "Please remember to do your walk after dinner later"), can reinforce consistency during this phase. The last stage, **Automaticity and Maintenance**, automates the behavior so that it requires little cognitive effort. At this point, external reminders should be faded to avoid dependency. Furthermore, habits that have become strong can now serve as anchors for new habits (habit stacking). This staged view emphasizes that habit formation requires adaptive support from guiding early motivation and contextual embedding to fostering automaticity.

Digital technologies can support the four stages. Apps can provide timely cues, personalized feedback, and adaptive reinforcement, helping users repeat desired behaviors until they become automatic (Reinsch, Weimann, et al., 2025). Moreover, they can bridge the intention-behavior gap by reducing the cognitive effort required to initiate actions (Stawarz et al. 2014, 2015). Features such as goal setting, gamification, and feedback help sustain motivation during the early, effortful stages of forming a habit, and adaptive reminders can facilitate long-term maintenance once behaviors become routine (Stark et al., 2023). Because these tools are ubiquitous, scalable, and capable of real-time adaptation, they are ideal for embedding behavioral science principles into daily life. To investigate how to support the habit formation stages best digitally, we conducted a literature review to identify literature that contributes to DPs for habit formation. We selected studies that (1) examined the psychological mechanisms underlying habit formation, (2) investigated digital interventions for behavior change involving habit components, and (3) formulated DPs for digital habit support.

Table 1 summarizes the literature identified in the review. Stark et al. (2023) propose a framework, that includes thirteen DPs related to one or more stages of the habit formation process. Most of the remaining papers focus on particular areas formulated within the framework of Stark et al. (2023). Stawarz et al. (2016) focus on reminders and contextual cues, providing empirical evidence that, in everyday life, individuals primarily rely on contextual cues, such as prior behavior or locations, to remember health-related behaviors. Habit formation is thus understood as a context-driven process, highlighting that digital reminders are more effective when anchored in natural situations rather than time. Pinder et al. (2016) examine context-sensitive implementation intentions ("if-then plans") and demonstrate that adaptive, non-intrusive digital triggers are crucial in bridging the intention-behavior gap, while contextual cues themselves remain only limitedly differentiated in the digital habit formation literature.

Schwarzer et al. (2018) focus on the sequential pathway from motivation to planning, self-monitoring, and eventual habitual behavior and investigate features such as information provision, personalization, and self-monitoring. An exception constitutes Zhu et al. (2024), who analyzed 41 digital interventions and identified 32 habit formation techniques. These techniques are more fine-grained and can help to detail the more abstract view persistent in Stark et al. (2023).

Literature	Focus / Method	Outcome	Key Design Implications
Stark et al. (2023)	Literature review and content Analysis of 57 apps	Formulation of 13 DPs (e.g., contextual cues, personalization, fading)	Include the 13 DPs for sustained habit formation
Stawarz et al. (2016)	3 surveys on real-world remembering strategies for medication adherence	Contextual cues (routines, objects, location) outperform technological reminders	Support contextual cues rather than alarms or tracking
Pinder et al. (2016)	Design probe (n=10) + survey (n=137) on context-aware “if-then” smartphone interventions	Users created health-related plans linking context cues (time, location, motion) with goal behaviors	Include configurable, context-triggered “if-then” plans with adaptive, non-intrusive, positively framed reminders
Zhu et al. (2024)	Systematic review of 41 studies on digital behavior change interventions for physical activity	Identified 32 habit formation techniques within a conceptual framework	Use the framework, including 32 DPs as a guideline to design digital behavior change interventions (DBCI)
Schwarzer et al. (2018)	Multicenter RCT (n = 1,564) testing psychological mechanisms	Sequential mediation: motivation, planning, self-monitoring, habit strength, physical activity	Include dynamic, and adaptive elements in particular to foster habit formation for women

Table 1. Literature on design principles for digital habit formation

Table 2 positions the identified studies within the four-staged habit formation process. The table maps DPs identified in the literature to the respective habit stages and indicates which studies support which principles. The mapping shows that Stage 1 is well represented in the literature, which formulates DPs such as *information provision*, *implementation intentions*, and *contextual specification*. In particular, Pinder et al. (2016) and Stawarz et al. (2016) examine the processes involved in articulating intentions and translating them into action. Stages 2 and 3, which share DPs, show a more heterogeneous pattern. While Zhu et al. (2024) emphasize a strong focus on user-oriented strategies, such as *personalization*, *motivation*, and *self-monitoring*, these aspects are less prominent in Pinder et al. (2016) and Schwarzer et al. (2018). Stages 3 and 4 of habit formation are comparatively underrepresented. Stage 3 requires a tendency toward automaticity, but this aspect is not sufficiently covered. For stage 4, only two DPs have been found. Taken together, the analysis reveals that research on habit formation is well developed in the area of habit initiation, yet habit strengthening and automaticity remain underexplored.

Design Principles	Stark et al. (2023)	Stawarz et al. (2016)	Pinder et al. (2016)	Zhu et al. (2024)	Schwarzer et al. (2018)	Stages
Information Provision	x	x		x	x	1
Implementation Intention*	x	x	x	x	O	1
Contextual Specification*	x	x	x	x		1
Avoid Information Overload	x					1
Habit Distinction	x		O			1
Flexible habit management	x			O		2-4
Trigger Events to maintain repetition	x	x	x	x		2
Personalization	x		x	x	x	1-4
Self-Monitoring	O	x		x	x	2-3
Social Interaction	O			x		2-3
Social Sharing	O	x		x		2-3
Praise Messages	O	x		O		2-3
Praise Rewards	O	x		x		2-3
Challenges and Levels	O			x		
Reminders to reinforce the implementation intention*	x	x	x	x		2-3
Tapered* Reminding	x	x	O	O		3
Fading Features*	x	x	O	O		4
Habit Stacking	x					4

Table 2. Summary of design principles across the four stages of habit formation. An “x” indicates that a study provides clear support for the principle, while “O” denotes partial or indirect evidence.

3 Method

This study employs a quantitative survey design, collecting empirical data from two key stakeholder groups: (1) scientists in the field of habit formation and digital health, and (2) end-users of habit formation apps. The survey instrument builds on a predefined construct space derived from prior design-oriented research on digital habit formation, which synthesized psychological theory and a content analysis of commercially available habit apps into a consolidated set of design principles (Stark et al., 2023). Building on this prior qualitative work, a quantitative design was deemed appropriate. Participants rated the importance of each DP on a five-point Likert scale ranging from 1 (“Not important at all”) to 5 (“Extremely important”), with intermediate categories of 2 (“Not important”), 3 (“Neither important nor unimportant”), and 4 (“Important”) (see Online Appendix, Table 3). These ratings capture participants’ subjective perceptions of importance and should not be interpreted as indicators of behavioral efficacy. A total of 208 participants initiated the survey. Of these, 36 identified themselves as scientists (22 completed; 57 % completion rate) and 172 as users (108 completed; 63 % completion rate). These completion rates are considered highly satisfactory for uncompensated online surveys targeting specific expert and practitioner populations (Shih & Xitao Fan, 2008). The researcher’s sample consisted of individuals who self-identified as scientists engaged in habit research, recruited through academic networks, including behavioral science conferences, habit research mailing lists, and direct outreach to authors of habit research. The user sample consisted of individuals who reported that they were currently attempting to form a habit or had already formed one. Eligibility was assessed at the beginning of the survey, and participants were instructed to focus on one personally relevant habit throughout the questionnaire. To further characterize the user sample, habit strength was assessed using the Self-Report Habit Index (SRHI); completed users (N = 108) showed a mean score of 3.19 (SD = 0.73; range: 1.00-4.92), indicating variation from less established to more established habits

(Verplanken & Orbell, 2003). Data collection was conducted between March 2024 and September 2025. All participants provided informed consent before survey completion and received no compensation for participation.

These findings were subsequently integrated into related research, specifically with the content analysis results of 57 commercially available habit apps reported by Stark et al. (2023), see Online Appendix Table 1. By combining these perspectives, this study identifies areas of convergence and divergence between theoretical, user-centered, and implementation-oriented views. This integrated approach enables an assessment of which DPs are prioritized across stakeholder groups, highlighting critical alignment gaps and priority areas for the evidence-based design of future habit-formation technologies. All data management and statistical analyses were performed using R (version 4.5.2). The analysis utilized the tidyverse package suite for data wrangling, reshape2 for data restructuring, ggplot2 for visualization, and base R functions for descriptive and inferential statistics. Additional visualizations were created in Tableau to enhance graphical clarity and comparability across stakeholder perspectives. The study followed institutional ethical guidelines and data protection regulations. All responses were collected anonymously and stored on secure university servers in accordance with GDPR requirements.

4 Results

The final sample comprised 22 scientists and 108 users (see online appendix Table 2). The demographic composition reflects natural differences between the groups. Users were predominantly younger (53.7% aged 18-24) than scientists (18.2% aged 18-24). This pronounced age difference raises the question of whether observed group differences reflect genuine stakeholder-role distinctions or are partly driven by age composition. We address this below through a robustness analysis. Educational attainment diverged markedly: 63.7% of scientists held graduate degrees (36.4% doctoral) versus 25.9% of users, reflecting the academic nature of the researcher sample. The gender distribution showed a male predominance among scientists (63.6%) and a more balanced representation among users (56.1% female). These demographic differences represent authentic stakeholder characteristics rather than sampling bias.

Analysis of importance ratings across 18 DPs revealed substantial overall alignment between scientists and users, with only four DPs demonstrating statistically significant differences in Mann-Whitney U tests (see Table 3). Notably, all four DPs with significant differences were rated higher by scientists than users, suggesting systematic patterns in how theoretical knowledge influences DP valuation. Effect sizes for all four significant features were small in magnitude (r range: $-.25$ to $-.30$), consistent with the exploratory nature of the study and reflecting meaningful but modest divergence between stakeholder groups.

Design Principle	Description	Mann-Whitney -U p	Scientist (mean)	User (mean)	r	95% CI
Information Provision	Providing educational content about the benefits of habits for decision-making.	0.116	3.59	3.87	.18	[-.06, .40]
Implementation Intention	Support for creating specific if-then plans that link contexts to behaviors.	0.207	3.36	3.01	-.14	[-.37, .10]
Contextual Cues	Specification of environmental triggers that prompt the execution of a habit.	0.016*	4.11	3.71	-.28	[-.48, -.04]
Information Overload	Avoid overwhelm by gradually disclosing information or categorizing goals.	0.346	3.92	3.72	-.11	[-.34, .13]
Habit Distinction	Differentiation between habit building/breaking.	0.247	3.5	3.72	.14	[-.11, .37]
Just-in-time Reminder	Delivery of notifications precisely when habits should be performed.	0.831	3.92	3.83	-.03	[-.28, .23]
Implementation Intention Reminder	Provision of pre-context notifications for reinforcement of if-then plans.	0.540	3.92	3.78	-.07	[-.32, .18]
Fading Reminder	Progressive reduction of reminder frequency as habits strengthen.	0.048*	3.75	3.25	-.25	[-.47, .00]
Self-Comparison	Visualization of current performance against personal history.	0.723	4.21	4.05	-.04	[-.29, .21]
Social Interaction	Comparison with other users via leaderboards and community features.	0.019*	3.25	2.55	-.30	[-.51, -.05]
Social Sharing	Facilitation of achievement and progress sharing between users.	0.019*	2.67	2.11	-.30	[-.51, -.05]
Challenges and Levels	Incorporation of challenges and difficulty levels for engagement maintenance.	0.162	4.18	3.8	-.17	[-.41, .09]
Praise Rewards	Provision of virtual rewards for achievements.	0.171	3.86	3.5	-.18	[-.42, .09]
Praise Messages	Delivery of motivational text messages.	0.680	3.73	3.59	-.05	[-.31, .21]
Fading Features	Reduction of gamification and external support during transition to automatic execution.	0.322	3.36	3.01	-.13	[-.37, .14]
Flexible Habit Management	Accommodation of habit pausing, skipping, or modification for changing circumstances.	0.288	3.91	4.11	.13	[-.13, .38]
Personalization	Customization of goals, reminders, interface, and behavioral goals to individual preferences.	0.328	4.14	4.08	-.13	[-.37, .14]
Habit Stacking	Facilitation of new habit anchoring to existing automatic routines for creating behavioral chains.	0.140	4.29	3.5	-.43	[-.78, .12]

Table 3. Comparison of scientists' and users' mean ratings for digital habit formation features. Asterisks () indicate statistically significant differences at $p < .05$. Importance ratings reflect subjective prioritization judgments on a five-point Likert scale (1 = not important at all, 5 = extremely important). r = rank-biserial correlation; negative values indicate scientists rated the feature higher than users. Effect size benchmarks: small $r \geq .10$, medium $r \geq .30$, large $r \geq .50$. All tests are exploratory; no multiplicity correction applied.*

Contextual Cues showed the strongest divergence ($p = 0.016$), with scientists rating this DP substantially higher ($M = 4.11$) than users ($M = 3.71$). Proportion analysis reveals this difference's magnitude: 78% of scientists (40.7% "extremely important", 37.0% "important") valued contextual cues highly compared to 66% of users (14.7% "extremely important", 51.7% "important"), while 10.3% of users rated this DP as "not important", a rating given by no scientists. This divergence likely reflects scientists' theoretical grounding in context-dependent automaticity versus users' potential perception of contextual specification as unnecessary complexity. Social DPs exhibited consistent patterns of divergence between researchers and users. Social Interaction differed significantly ($p = 0.019$) between scientists ($M = 3.25$) and users ($M = 2.55$). Specifically, 51.9% of users rated this DP as "not important" or "not important at all," compared to only 12.5% of scientists. Social Sharing showed similar patterns ($p = 0.019$; scientists, $M = 2.67$, users, $M = 2.11$), with 67.6% of users assigning low importance ratings compared to 37.5% of scientists. These differences suggest that scientists recognize the potential theoretical value in social mechanisms, whereas users view them as peripheral to the habit formation process. Fading Reminders achieved marginal significance ($p = 0.048$), with scientists ($M = 3.75$) valuing the reduction of reminders more than users ($M = 3.25$). This finding aligns with theoretical principles of transitioning from external to internal regulation, which may not be intuitive to end users.

While not reaching statistical significance, three DPs showed higher mean ratings from users than scientists, revealing areas where practical experience may diverge from theoretical expectations. Information Provision was rated higher by users ($M=3.87$) than scientists ($M=3.59$, $p=0.116$), as was Habit Distinction (users $M=3.72$, scientists $M=3.50$, $p=0.247$) and Flexible Habit Management (users $M=4.11$, scientists $M=3.91$, $p=0.288$). These preferences suggest users value practical flexibility and clear information more than scientists. Core tracking and reminder DPs demonstrated strong consensus between groups, with Self-Comparison receiving consistently high ratings (scientists, $M = 4.21$; users, $M = 4.05$, $p = 0.723$) and Just-in-Time Reminders showing near-identical importance (scientists, $M = 3.92$; users, $M = 3.83$, $p = 0.831$), indicating shared recognition of habit support mechanisms. Both groups demonstrated sufficient engagement with the survey to provide valid responses, with no patterns of straight-lining or implausible response combinations that would suggest inattentive responding. To confirm that these group differences reflect stakeholder-role distinctions rather than age composition, we conducted a robustness analysis restricting both groups to participants aged 25 and older (users: $n = 50$; scientists: $n = 18$). All four significant findings held. Contextual Cues remained significant, though with a slightly attenuated effect ($p = .044$), while Social Interaction ($p = .009$), Social Sharing ($p = .009$), and Fading Reminders ($p = .034$) all grew stronger after the restriction, suggesting that the younger user subgroup had partially suppressed these differences rather than inflated them.

5 Discussion

Our investigation shows that users and scientists share agreement on several DPs, particularly those not exclusively tied to the theoretical core of habit formation but supportive of day-to-day behavior execution and habit management (see Figure 1). Self-Comparison, Just-in-Time Reminders, Flexible Habit Management, and Personalization received the highest importance ratings across both groups, indicating broad consensus on features that make habit performance more manageable, transparent, and adaptable to individual needs. This convergence suggests that stakeholders align most strongly on pragmatic, user-facing functions that directly facilitate engagement and everyday behavioral consistency. At the same time, users attributed slightly higher value to practical elements such as Information Provision, Habit Distinction, and Flexible Habit Management, suggesting that end users prioritise clarity, flexibility, and straightforward guidance when forming habits.

Contextual Cues, Social Interaction, Social Sharing, and Fading Reminders differed significantly between groups. Contextual cues and fading reminders are foundational to habit formation. Contextual cues help ensure that behaviors become embedded within stable daily routines, while fading reminders support the persistence of behaviors once technological scaffolding is reduced or removed. These mechanisms are essential for establishing strong, context-driven automaticity, which explains why scientists place greater emphasis on them than users (Verplanken, 2018; Zhu, Long, Wei, et al., 2024).

More unexpected are the differences observed for social interaction and social sharing, since these features have not traditionally been considered central to habit formation. One possible explanation is that social mechanisms have long been recognised as influential in broader behavior change processes through accountability, commitment, and social reinforcement (Zhu, Long, Wei, et al., 2024). Scientists may therefore attribute greater potential to these features as indirect facilitators of habit development, even if users view them as less relevant to their individual habit journeys.

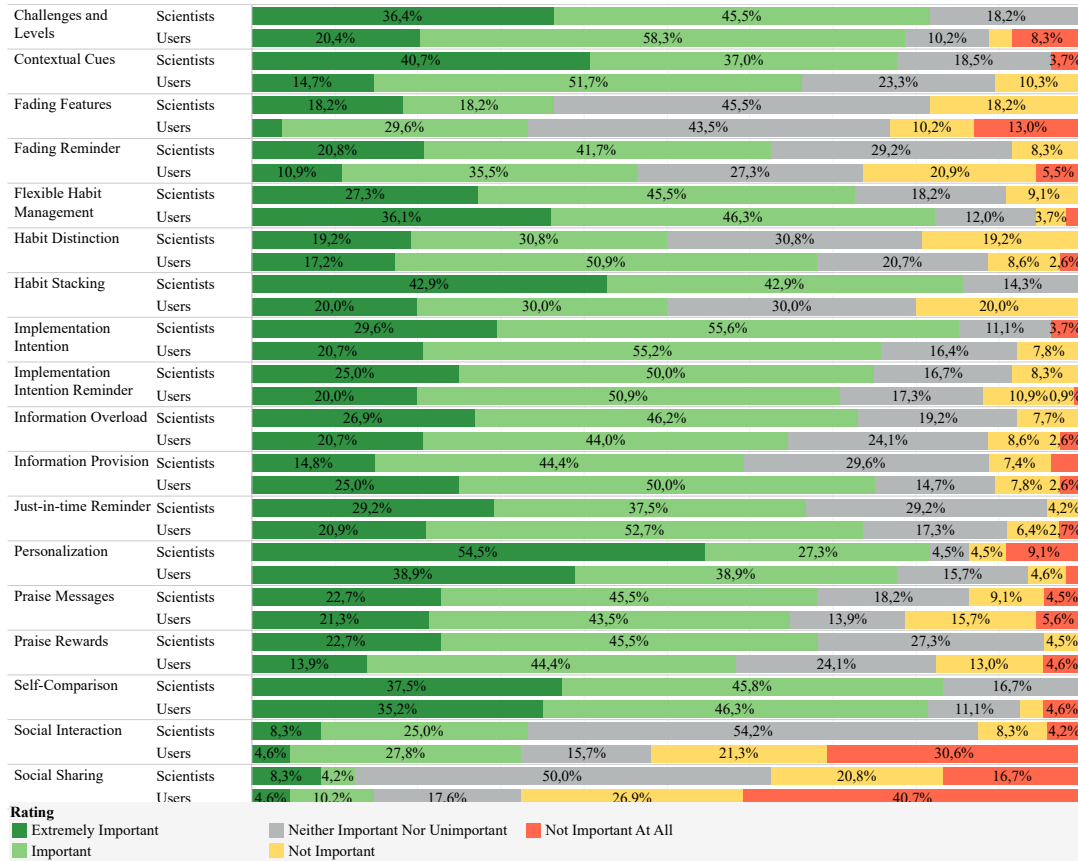


Figure 1. Feature importance ratings across scientists and users

While these stakeholder perspectives reveal important areas of consensus and divergence regarding which DPs should guide habit formation apps, the critical question remains: How do these theoretical and user-centered priorities translate into commercial practice? To address this gap between perceived importance and market reality, we now examine implementation data from Stark et al.'s (2023) systematic content analysis of 57 habit formation apps. Figure 1 in the online appendix presents implementation scores for each design principle. Just-in-Time Reminders ($M = 1.89$ out of 2.0) and Self-Comparison ($M = 1.81$) achieved near-universal implementation, while gamification features, such as Challenges, Levels, and Praise Rewards (both $M = 1.49$), were similarly prevalent. Flexible Habit Management ($M = 1.35$) was present in most apps. However, theoretically critical design principles were virtually absent: Fading Reminders and Fading Features both scored merely 0.0175 (less than 1% of apps), despite moderate-to-high stakeholder importance ratings. These fading mechanisms are fundamental for transitioning from external support to automaticity. Implementation Intentions, rated highly important by both scientists ($M = 4.07$) and users ($M = 3.89$), achieved minimal implementation ($M = 0.316$). Similarly, Contextual Cues, despite high scientist ratings ($M = 4.11$), achieved only moderate implementation ($M = 0.842$), with apps defaulting to simple time-based reminders rather than robust context specification.

To better understand stakeholder priorities and real-world application design, we compare perceived importance with actual implementation levels in an implementation-importance matrix (see Figure 2). The matrix positions each DP according to its perceived importance and degree of implementation,

creating a portfolio perspective that reveals four distinct quadrants. These quadrants highlight areas of **Aligned Priorities** (top right), **Opportunity Areas** (bottom right), **Overinvested Principles** (top left), and **Low-Value Principles** (bottom left), offering a structured view of where scientist and user expectations, as well as market practice, converge or diverge. The size of each point reflects the absolute difference between scientists' and users' importance ratings, allowing the matrix to visualize the magnitude of stakeholder disagreement.

Aligned Priorities (High Importance / High Implementation; blue). DPs in the upper-right quadrant represent areas where stakeholder expectations and market practice are closely aligned. Based on the averaged importance ratings of scientists and users, Self-Comparison, Just-in-Time Reminders, Flexible Habit Management, and Personalization all emerge as highly valued features that also show strong implementation in current habit apps. This convergence indicates that developers have successfully prioritized principles that offer immediate practical value and are intuitively understood by users. These features function as established best practices in digital habit support because they provide explicit, actionable scaffolding that helps users initiate and maintain behaviors, monitor their progress, and adapt habitual routines to changing circumstances. Their widespread implementation reflects both their usability and their perceived utility, reinforcing their status as foundational components of contemporary habit formation technologies.

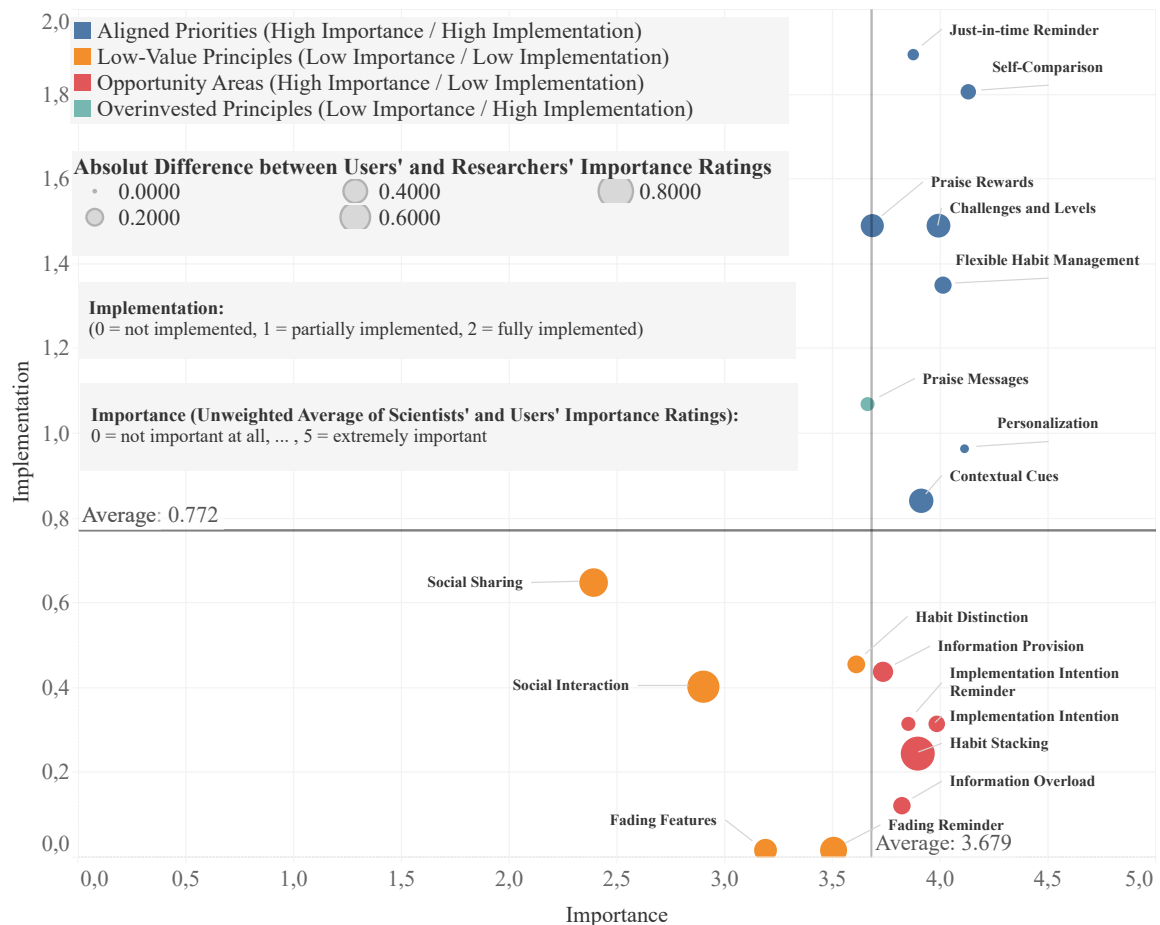


Figure 2. Implementation–importance matrix mapping the perceived importance and actual implementation of 18 design principles.

Opportunity Areas (High Importance / Low Implementation; red). DPs in the bottom-right quadrant represent features that are highly valued by scientists and users yet remain largely unimplemented in current habit apps. The most prominent examples include Implementation Intentions, Information Overload Avoidance, Implementation Intention Reminders, and Habit Stacking. These principles capture mechanisms that are central to habit formation, including the creation of strong context-behavior

links, the formulation and reinforcement of implementation intentions, and the gradual cognitive offloading required for long-term automaticity. Their consistently high importance ratings, averaged across stakeholder groups, underscore a clear recognition of their potential value. However, their minimal implementation suggests that developers either underestimate their relevance or face barriers in translating these theoretically grounded mechanisms into practical design solutions. This quadrant highlights the most critical strategic opportunities for innovation, pointing to areas where integrating habit formation principles more explicitly into app design could generate substantial gains in habit formation effectiveness.

Overinvested Principles (Low Importance / High Implementation; green). DPs in the top-left quadrant represent features that receive comparatively low importance ratings from scientists and users yet are widely implemented in commercial habit apps. Praise Messages provide the clearest example of this pattern. Their strong presence in apps suggests that developers prioritize engagement-oriented features that align with common product design conventions, even when the value of these features for habit formation is unclear. This mismatch indicates a potential misallocation of development resources toward capabilities that may enhance short-term motivation but contribute little to the underlying mechanisms necessary for building stable, automatic behaviors.

Low-Value Principles (Low Importance / Low Implementation; orange). DPs in the bottom-left quadrant represent features that receive low importance ratings from scientists and users and show minimal implementation in commercial habit apps. This group encompasses social mechanisms, including Social Interaction and Social Sharing. Although these principles differ in their theoretical relevance, their joint position in this quadrant suggests that they are perceived as less central to supporting digital habits. Interestingly, also fading features can be found in this quadrant, which are discussed as central to habit formation (Stark et al., 2023; Stawarz et al., 2025)

While Figure 2 provides a consolidated view of how importance and implementation align at an aggregate level, habit formation is not a uniform process. Figure 3, therefore, breaks down the analysis by habit formation stage, showing how each DP performs within the four stages of habit formation. In this stage-specific visualization, circles represent user ratings and crosses represent scientist ratings, enabling the examination of differences between stakeholder perspectives within each stage. Despite its complexity, Figure 3 offers a more nuanced understanding of where alignment emerges, where divergence persists, and where the most critical gaps occur along the progression from initial intention to long-term automaticity. In the following, we discuss selected DPs for each stage.

The **Intention Stage (top/left)** marks the beginning of the habit formation process, where individuals decide on a behavior and form an initial commitment to perform it. At this point, apps are expected to help users clarify their goals and determine when and where the behavior should occur, laying the foundation for later cue-behavior associations. Two DPs are fundamental here: contextual cues and implementation intentions, which jointly establish the mechanisms through which intentions become reliably enacted (Pinder et al., 2016). Contextual cues show the strongest divergence between stakeholder groups ($p = 0.016$). Scientists rate them considerably higher (4.11) than users (3.71), likely because scientists conceptualize habits as context-dependent responses, whereas users tend to think in terms of time-based routines. This difference is reflected in practice: although contextual cues achieve moderate implementation (0.842), most apps rely on simple time-based notifications rather than supporting richer contextual embedding, such as linking behaviors to existing routines (e.g., brushing one's teeth after breakfast) or stable environmental triggers (e.g., putting on running shoes after arriving home). Yet, without such embedding, apps fail to create the cue-behavior links required for automaticity (Stawarz et al., 2016; Zhu, Long, Wei, et al., 2024). Implementation intentions serve a complementary function, translating intentions into actionable "if-then" plans that bind behaviors to specific contextual cues (e.g., "if I finish breakfast, then I will take a short walk"). Despite high importance ratings from both groups (scientists 4.07, users 3.89), implementation intentions are rarely implemented (0.316). Many apps allow general goal setting but do not guide users through the cognitive planning necessary to anchor behaviors in concrete contexts (Gollwitzer, 1999). Taken together, contextual cues and implementation intentions form an integrated system for habit initiation; when either component is missing, behavior remains externally regulated rather than becoming contextually grounded.

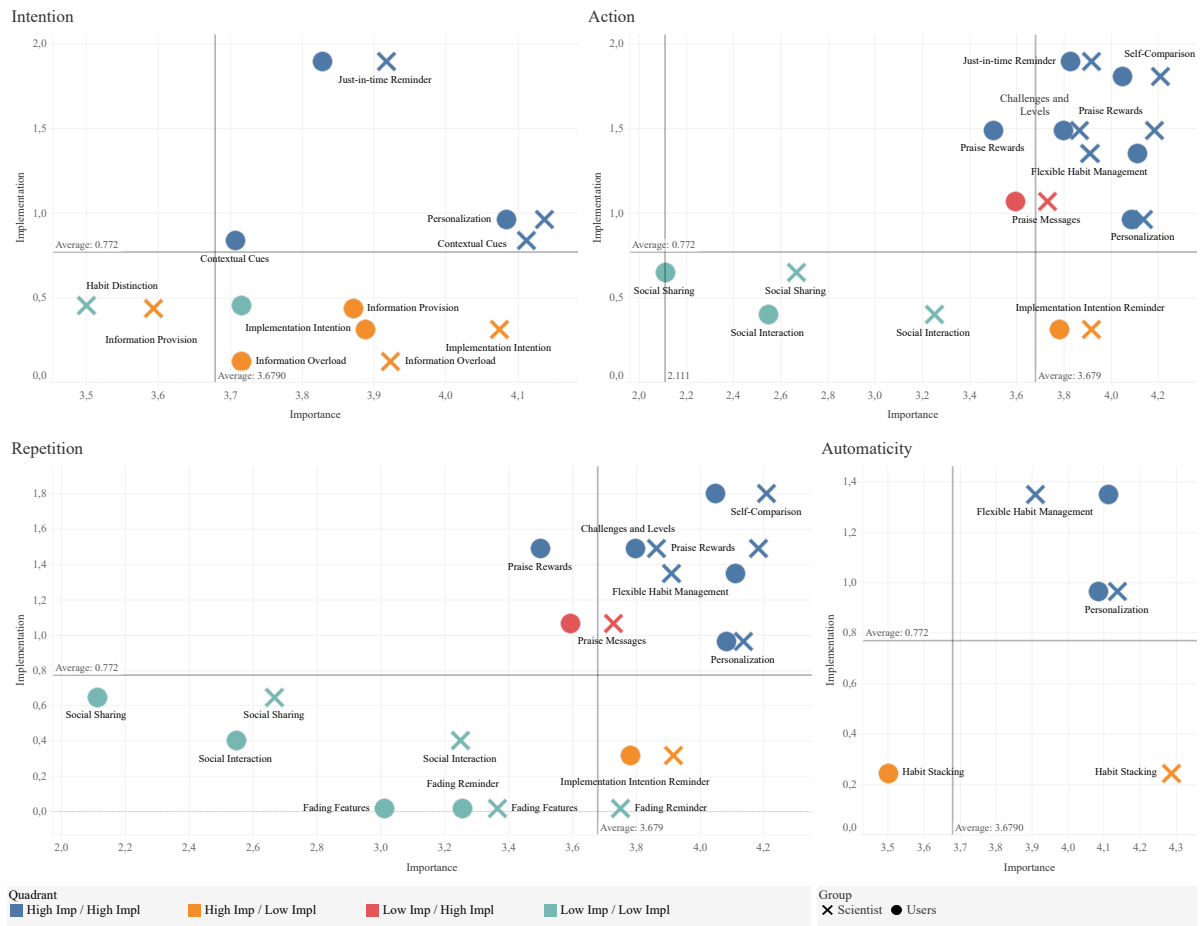


Figure 3. Stage-specific implementation–importance matrices showing how scientists and users evaluate and experience 18 design principles across the four stages of habit formation (Intention, Action, Repetition, Automaticity); Stage 1 – Intention; Stage 2 – Action; Stage 3 – Repetition; Stage 4 – Automaticity.

Beyond these core mechanisms, information provision and the avoidance of information overload also influence the intention stage. Users require sufficient and well-structured information to select a meaningful behavior, yet information overload avoidance remains almost absent in current apps (0.123), despite moderate importance ratings. Furthermore, reminders are rated as necessary by both groups (scientists: 3.92, users: 3.83), although they play a functional role only once behavioral performance begins. Their widespread implementation (1.89) suggests that developers prioritize easily deployable time-based reminders early on, even though such notifications offer limited support for the cognitive processes that define habit initiation and may inadvertently reinforce external regulation rather than fostering contextual embedding.

The **Action Stage** marks the transition from intention to the first instances of behavior performance. At this point, apps are expected to help users translate their implementation intentions into concrete action, support early execution, and reduce barriers to getting started. Several DPs show strong stakeholder–implementation alignment in this stage, including Self-Comparison, Just-in-Time Reminders, Flexible Habit Management, and Personalization, all of which receive high importance ratings and strong implementation. These features enable users to initiate behaviors, monitor early progress, and adjust routines to accommodate daily life. A misalignment emerges for Social Features. Both Social Interaction and Social Sharing receive low importance ratings from users (2.55 and 2.11), with a majority deeming

Social Sharing unimportant (67.6%). Yet, implementation remains moderate (0.649), likely reflecting platform logics that favor virality over evidence-based support for behavior change.

The **Repetition Stage** marks the shift from initial behavior performance toward the gradual emergence of automaticity. At this point, the goal is no longer merely getting users to act but helping them repeat the behavior consistently in stable contexts, so that it begins to transition from deliberate effort to cue-driven execution. Digital support should therefore focus on strengthening context-behavior associations and reducing external regulation as users gain confidence and behavioral stability (Stawarz et al., 2016). However, the Repetition Stage reveals that apps often fail to provide the necessary mechanisms for this transition. Fading Reminders, which are theoretically essential for reducing external prompting and allowing contextual cues to take over, show statistically significant stakeholder divergence ($p = 0.048$) and remain virtually absent in current apps (0.0175). A similar pattern appears for Fading Features (0.0175), despite moderate importance ratings from both groups (scientists 3.36, users 3.01). Technical barriers cannot explain this absence: simple algorithms could gradually reduce reminder frequency or scale back feedback and gamification elements as behavioral consistency increases (Reinsch, Weimann, et al., 2025). Instead, it reflects a persistent design bias toward engagement optimization, where continuous prompting, feedback, and rewards are maintained even when theoretical models call for their reduction. Implementation Intention Reminders show a similar mismatch. Although theoretically central for strengthening the link between contextual cues and behavioral execution during repeated practice, they receive high importance ratings (scientists: 3.92, users: 3.78) but remain minimally implemented (0.316). This gap may partly stem from the deficiencies identified in the Intention Stage: apps cannot reinforce if-then plans during repetition if those plans were never scaffolded correctly in the first place. The Repetition Stage, therefore, exposes cascading failures across stages, where missing support in early planning undermines later attempts to automate behavior. One notable exception is Flexible Habit Management, which shows strong implementation (1.35) and even higher importance among users than scientists (4.11 vs. 3.91). This suggests that users intuitively recognize that rigid tracking systems can derail habit development when disruptions occur.

The **Automaticity Stage** represents the culmination of the patterns identified across previous stages. Habit Stacking, the primary DP mapped to this stage, achieved minimal implementation (0.246) despite receiving the highest importance rating from researchers across all features ($M = 4.29$) and moderate user endorsement ($M = 3.50$). This gap reflects a lost opportunity; established habits provide the strongest contextual anchors for new behavior development, offering what theory identifies as optimal integration points for behavior chaining. The minimal implementation likely stems from technical challenges in detecting users' existing habits and designing interfaces that support complex behavioral sequences. Yet, these challenges do not fully explain the absence, given that a simple user self-report could identify anchor habits. More fundamentally, the Automaticity Stage reveals that apps lack any awareness of when habits have achieved automaticity. No app demonstrated the capability to detect automaticity markers, whether through consistency metrics exceeding, reduced response latency patterns, increased context stability, or validated self-report measures of automaticity strength (Stark et al., 2023). Instead, apps maintain identical interaction patterns across habit development stages, treating day-one users the same as those with year-long streaks. This absence of stage-aware adaptation extends the pattern established in previous stages: just as apps fail to implement contextual specification at **Intention**, maintain external regulation at **Action**, and avoid fading mechanisms at **Repetition**, they completely neglect to recognize or respond to the achievement of **Automaticity**.

Taken together, the findings reveal a consistent pattern across importance ratings, implementation status, and stage-specific analysis. First, users and scientists broadly agree on DPs that support everyday execution, such as Self-Comparison, Just-in-Time Reminders, Flexible Habit Management, and Personalization, indicating that both groups prioritize pragmatic support mechanisms that make behavior performance easier and more manageable. Second, the biggest differences occur for theoretically central mechanisms, most notably Contextual Cues and Fading Reminders, which scientists value more highly but are underappreciated by users and rarely implemented. Third, by integrating stakeholder priorities with implementation data, the portfolio analysis highlights a substantial misalignment between theoretical requirements and existing app design: features critical for habit

formation (e.g., Implementation Intentions, Contextual-Cues, Fading Reminders) remain largely unrealized, while engagement-oriented features are often overemphasized. The stage-based analysis reinforces this conclusion by showing how these gaps accumulate over time: apps support early action but fail to cultivate the contextual embedding, repetition dynamics, and withdrawal of external support needed for behaviors to become truly automatic. Across all stages, the core insight remains the same: current habit apps help users start but rarely allow them to develop genuine habits. Habit formation is a marathon, not a sprint.

6 Limitations

While this study offers an integrated perspective on DP prioritization and real-world implementation, several limitations warrant consideration. The primary constraint concerns the reliance on secondary content analysis data from Stark et al. (2023). Their analysis captures app features at a specific point in time; however, the rapid evolution of mobile apps means that current implementations may differ from those documented. More critically, mapping Stark et al.'s coding taxonomy to our survey-based DPs required interpretative alignment that may not fully capture implementation nuances. Furthermore, the ternary coding scheme (0-1-2) reflects feature presence rather than quality, user engagement, or actual usage patterns. An app might achieve full implementation scores for Social Sharing (2.0) despite users never activating these features, while partially implemented Self-Comparison capabilities (1.0) might drive daily engagement and behavioral reinforcement. Without complementary usage analytics or effectiveness metrics, implementation scores provide limited insight into which features genuinely facilitate habit formation versus merely existing as interface elements. This disconnection between presence and utilization represents a limitation when assessing theory-practice alignment, as theoretically sound features implemented poorly or positioned in an inaccessible manner may appear aligned while failing to support actual habit formation. Additionally, the use of identical instruments across stakeholder groups with differing expertise raises concerns about construct validity. While standardized explanations were provided for each principle, future studies should complement quantitative prioritization with qualitative methods to examine whether observed divergences reflect genuine differences in valuation or differences in construct interpretation. Despite these constraints, this study provides the first systematic triangulation of stakeholder priorities with commercial implementation, establishing a foundation for future experimental validation of DPs effectiveness in supporting genuine behavioral automaticity.

7 Future Research

The stage-based patterns identified in this study illuminate several avenues for future research. The systematic amplification of theory-practice gaps across habit formation stages, from moderate alignment in early stages to profound disconnects at the automaticity stage, warrants a longitudinal investigation examining how these implementation deficits affect actual habit formation outcomes. Future studies should empirically test whether apps with stronger implementation of late-stage features (fading mechanisms, automaticity detection) produce more robust habit formation compared to current engagement-optimized designs. Experimental comparisons that manipulate the presence or absence of theoretically critical features could establish relationships between DPs and the development of habit strength, thereby moving beyond the correlational patterns documented here.

The consistent divergence between researcher and user importance ratings across theory-driven features highlights the urgent need for research on knowledge translation mechanisms in digital health design. Future work should investigate optimal strategies for communicating complex behavioral science concepts to end users, potentially through interactive onboarding sequences. These progressive disclosures of theoretical rationales, or gamified tutorials, demonstrate why initially counterintuitive features, such as gradual support reduction or detailed context specification, ultimately facilitate behavior automaticity (Zhu, Long, Wang, et al., 2024). User experience research can identify interface metaphors and interaction patterns that make sophisticated habit formation mechanisms accessible to users without a background in behavioral science. Additionally, research should examine whether users

who understand the theoretical basis for design features demonstrate improved habit formation outcomes compared to those using apps without such education.

Most critically, future research must address whether current apps contribute meaningfully to lasting behavior change or merely maintain behaviors through continued external regulation. Longitudinal studies tracking users after app discontinuation, whether voluntary or forced through technical issues, could definitively test whether behaviors persist as genuine habits or collapse without technological support. Such research should employ ecological momentary assessment to capture real-time behavioral patterns, use objective measures where feasible (e.g., accelerometry for physical activity, financial records for saving habits), and extend follow-up periods beyond the 66-day average often cited for habit formation (Singh et al., 2024). Randomized controlled trials comparing apps with strong implementation of fading mechanisms versus standard constant-support designs would provide evidence for the importance of progressive autonomy development. These studies should also examine individual differences in optimal support trajectories, as some users may require longer periods of external regulation. In contrast, others benefit from a rapid transition to self-regulation (Reinsch, Weimann, et al., 2025). Understanding these individual differences could inform adaptive algorithms that personalize fading schedules based on user characteristics and behavioral patterns, representing a sophisticated integration of theory, technology, and personalization that current apps lack entirely (Reinsch et al., 2024, 2026).

8 Online Appendix

To enhance the clarity and coherence of the study, additional supporting materials are provided in the online appendix, accessible via the following link: <https://t1p.de/c3smi>

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